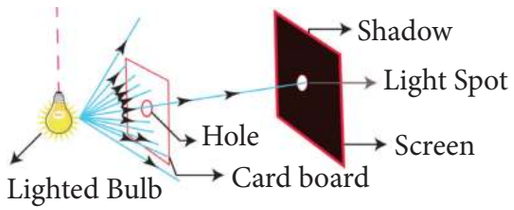
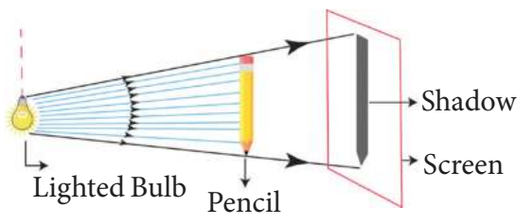
3. It cannot be determined the characteristics of an object by its shadow.

4. The shadow will be always darker, whatever may be the colour of light rays.

5. Light source, opaque object are shadow all are in a straight line.

6. The size of shadow depends upon the distance between light source and object and the distance between object and the screen.

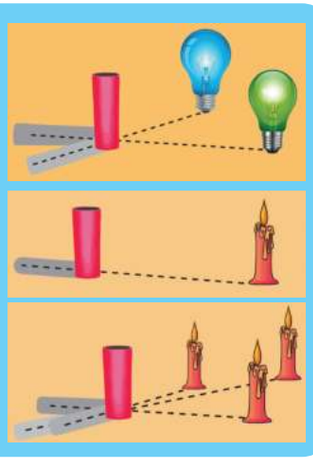
ACTIVITY 5

Arrangement	Activity	Observation	You Learn
	Place a lighted bulb in front of a rectangular card board with a hole at the center	A shadow with a spot light appears in the screen.	Light rays are passing only through the hole and are not allowed by the remaining part of card board
	Place a pencil in the path of light ray coming from a bulb	A shadow of pencil appears in the screen	The size of the shadow is proportional to the size of the opaque objects.

ACTIVITY 6

Requirement : A white screen, a cylindrical opaque object and three bulbs in different sizes.

Use the three different size lamps and examine the umbra and penumbra formed. Keep the distance between the lamp and the cylinder, cylinder and the screen same. As the size of the lamps grow smaller, the umbra region begins to enlarge. If the size of the lamp is a point, then there will be no penumbra. There would be only umbra shadow. Can you tell what the reason is for that?



Eclipses

An eclipse is an incident, when any astronomical object is partially or fully obscured due to the placement of another astronomical object in the presence of light. Thus, solar and lunar eclipses are occurring that are due to the property of light known as the rectilinear propagation of light.

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